

ATENOLOL

(TENORMIN)

Antihypertensive | Beta-1 Selective Adrenergic Blocker

Pharma Talk

1 MECHANISM OF ACTION (MOA) & TARGET

- **TARGET:** Beta-1 adrenergic receptors
- **MOA:** Blocks beta-1 receptors in heart → ↓ heart rate & contractility → ↓ cardiac output & BP
- **Analogy:** A biological speed governor slowing the heart's engine.



2 FORMS & STRENGTHS

- Oral (PO) tablets
- Strengths: 25 mg, 50 mg, 100 mg
- White, round tablets; imprinted as per manufacturer.



3 INDICATIONS (FDA-APPROVED & MAJOR OFF-LABEL)



FDA-APPROVED

- Hypertension (HTN)
- Chronic stable angina pectoris
- Hemodynamically stable acute MI

OFF-LABEL USES

- Atrial fibrillation rate control
- Migraine prophylaxis
- Essential tremor

4 DOSAGE & ADMINISTRATION

A) ADULT DOSE

- Standard: 50 mg PO daily
- Max: 100 mg daily
- Maintenance only
- Route: Oral
- Take consistently daily

B) PEDIATRIC DOSE

PEDIATRIC USE — Safety/efficacy not established.

C) ADMINISTRATION

- Route: PO
- Take with or without food
- Take at same time daily

D) ADJUSTMENTS

- Renal (CrCl based):
  15–35 mL/min: Max 50 mg/day
  <15 mL/min: Max 25 mg/day
- Hepatic impairment: None

5 PHARMACOKINETICS & DYNAMICS (PK/PD)



Onset: 1 hour

Half-life ($T_{1/2}$): 6–7 hours

Metabolism/Excretion: Renal excretion unchanged

6 ADVERSE EFFECTS & TOXICITY



COMMON

- Bradycardia
- Fatigue
- Dizziness
- Cold extremities

SEVERE (HIGH RISK)

- Bronchospasm
- Heart block
- Exacerbation of heart failure

7 CONTRAINDICATIONS & INTERACTIONS

ABSOLUTE CONTRAINDICATIONS

- Cardiogenic shock
- Decompensated heart failure
- Severe bradycardia
- 2nd or 3rd degree AV block

BLACK BOX WARNING

Abrupt withdrawal causes ischemia/MI.

MAJOR INTERACTION



Synergistic depression → Severe bradycardia

8 SPECIAL POPULATIONS (PREGNANCY & LACTATION)

Pregnancy: Fetal growth restriction risk.

Lactation: Excreted in breast milk; monitor infant for bradycardia.

Geriatric: Start low; monitor HR & BP closely.



9 CLINICAL PEARLS



- Cardioselectivity is dose-dependent; beta-2 blockade at high doses → bronchospasm risk in asthmatics.
- Masks tachycardia from hypoglycemia in diabetics.
- Taper over 1–2 weeks before discontinuation to prevent rebound angina.
- Monitor HR and BP before administration.

10 GLOSSARY OF ABBREVIATIONS



- **BP** = Blood Pressure
- **HTN** = Hypertension
- **MI** = Myocardial Infarction
- **CrCl** = Creatinine Clearance
- **mL/min** = Milliliters per Minute
- **HF** = Heart Failure
- **AV** = Atrioventricular
- **DDI** = Drug-Drug Interaction
- **CCBs** = Calcium Channel Blockers
- **CI** = Contraindication
- **Non-DHP** = Non-Dihydropyridine
- **HR** = Heart Rate